Recovered drift and diffusion models

SSISDE experiment results

20 October 2026

Final Stochastic SINDy fits using noise recovered from each drift method: $dX_t = \widehat{\mu}(X_t)\,dt + \widehat{\sigma}(X_t)\,dW_t$.

Method	Recovered drift $\widehat{\mu}(x)$	Recovered diffusion $\widehat{\sigma}(x)$	α	ρ
Black–Scholes				
Method 1	$0.503773\,x$	$0.00323797 + 0.295761\,x$	3.8746751×10^{-4}	0.85
Method 2	$0.501688\,x$	$0.00318241 + 0.295818\,x$	3.8746751×10^{-4}	0.85
ODE2	$0.501855\,x$	$0.00318859 + 0.295811\,x$	3.8746751×10^{-4}	0.85
Constant coefficients				
Method 1	$0.499733 + 0.000265495\,x$	$1.0002 - 7.42992 \times 10^{-8}\,x$	1.7190722×10^{-4}	0.3
Method 2	0.5	1.0002	1.3111339×10^{-3}	0.3
ODE2	0.5	1.0002	1.3111339×10^{-3}	0.3
Quadratic coefficients				
Method 1	$0.620512\,x + 0.35905\,x^2$	$-0.0177782 + 0.535985\,x + 0.0842435\,x^2$	8.7332616×10^{-4}	0.5
Method 2	$1.02213\,x$	$-0.0199718 + 0.540087\,x + 0.0823874\,x^2$	8.7332616×10^{-4}	0.4
ODE2	$0.427313\,x + 0.450974\,x^2$	$-0.0201858 + 0.540204\,x + 0.0824996\,x^2$	8.7332616×10^{-4}	0.4
Ornstein–Uhlenbeck				
Method 1	$-0.500311 + 0.000313962\,x$	$0.299011 - 1.72317 \times 10^{-7}\,x$	8.7332616×10^{-4}	0.4
Method 2	-0.5	0.29901	8.7332616×10^{-4}	0.4
ODE2	$-0.5\,x + 1.4265 \times 10^{-7}\,x^2$	$0.29901 + 5.9507 \times 10^{-10}\,x$	1.7190722×10^{-4}	0.4

Parameters. α is the ElasticNet regularization strength; ρ is its L1 mixing ratio. Each pair applies to the joint drift–diffusion fit and was selected by seven-fold cross-validation with the one-standard-error rule. All four experiments used 100,000 simulated samples.

Precision. Polynomial coefficients retain the precision stored in the experiment results, including small nonzero terms. α is displayed to eight significant digits; its full stored value appears in the LaTeX source comments. $\widehat{\sigma}$ is the diffusion amplitude, not its square.

Source. `outputs/<case>/ssisde_comparison.json`, where `<case>` is `black_scholes`, `constant`, `quadratic`, or `ou`.